

REDRAG Tools — Tool Creation Guide

How to Build a New Tool in 9 Steps

Phase 8 Framework | Zero Coding Knowledge Required

OVERVIEW

Goal: Create a new calculator/tool for tools.redrag.in

Time Required: 30-60 minutes per tool

Skills Needed: Copy-paste only. No coding knowledge required.

What You Do: Edit text values in pre-made files.

What AI Does: Generate the calculation logic (if needed).

HOW THE FRAMEWORK WORKS

Every tool is a self-contained folder:

```
YOUR_TOOL_SLUG/  
├─ index.html    ← SEO, meta tags, page structure  
├─ tool.css      ← Tool-specific styles (<150 lines)  
├─ tool.js       ← Calculation logic + schemas  
└─ data.json     ← Tool data (optional)
```

Shared Framework (already built, never touch):

```
shared/  
├─ css/          ← core.css, components.css, shell.css  
├─ js/           ← core.js, components.js, seo.js  
├─ html/         ← header.html, footer.html  
└─ data/         ← tools-registry.json
```

Key Principle: Each tool folder is independent. Copy template → Edit → Done.

STEP-BY-STEP PROCESS

STEP 1: Copy the Template Folder

1. GitHub repo open karo: github.com/aapka-username/tools.redrag.in
2. `template/` folder dhoondo
3. “Add file” → “Create new file” nahi — folder copy karna hai
4. GitHub pe direct folder copy nahi hota, toh yeh karo:
 - **Option A:** Local computer pe `template` folder copy karo, rename karo, fir GitHub Desktop ya direct upload karo
 - **Option B:** Har file individually create karo (niche bataya hai)

Folder name kya rakhein?

Short, lowercase, hyphen-separated: -  vram

-  vram
-  ram-latency
-  pc-build
-  VRAM Calculator (spaces nahi)
-  RamLatency (camelCase nahi)

STEP 2: Create the Folder on GitHub

GitHub pe new folder banane ka tarika:

1. “Add file” → “Create new file”
2. File name mein folder path likho:

```
vram/index.html
```

(Isse vram/ folder automatically ban jayega)

3. Template ka index.html content copy-paste karo
4. “Commit new file”
5. Same process for:

```
vram/tool.css
vram/tool.js
vram/data.json
```

STEP 3: Edit index.html (SEO & Content)

Find and Replace these placeholders:

Placeholder	Replace With	Example
TOOL_NAME	Tool ka naam	VRAM Calculator
TOOL_TAGLINE	Short catchy line	How Much VRAM Do You Need?
TOOL_SLUG	Folder name	vram
TOOL_DESCRIPTION	1-2 line description	Calculate GPU memory requirements for AI models
TOOL_KEYWORDS	Comma-separated keywords	VRAM calculator, GPU memory, AI models
TOOL_SHORT_DESCRIPTION	Hero section text	Find out if your GPU can run Llama 3

Table 1 – continued

Placeholder	Replace With	Example
TOOL_LONG_DESCRIPTION	SEO article text	VRAM (Video RAM) is critical for...

Important paths to update:

```
<!-- Canonical URL -->
<link rel="canonical" href="https://tools.redrag.in/vram/">

<!-- OG Image -->
<meta property="og:image" content="https://tools.redrag.in/assets/images/og-vram.jpg">

<!-- Twitter Image -->
<meta name="twitter:image" content="https://tools.redrag.in/assets/images/og-vram.jpg">
```

Schema mein update karo:

```
<!-- BreadcrumbList mein -->
"name": "VRAM Calculator"
"item": "https://tools.redrag.in/vram/"
```

STEP 4: Edit tool.css (Styling)

Most tools don't need CSS changes.

Template CSS already covers common patterns.

Only edit if:

- Aapko special input chahiye (range slider, color picker, etc.)
- Result ka display alag tarike se chahiye
- Mobile pe kuch break ho raha hai

Rule: CSS 150 lines se zyada nahi honi chahiye.

Agar zyada chahiye toh shared framework mein add karo (next AI se bolo).

STEP 5: Edit tool.js (Calculation Logic)

Yeh sabse important step hai.

3 cheezein edit karni hain:

5A: CONFIG Object (Top of file)

```
const CONFIG = {
  name: 'VRAM Calculator',           // Tool ka display name
  slug: 'vram',                     // Folder name
  version: '2.0.0',
  dataUrl: 'data.json',             // Data file ka path
```

```
// Input fields define karo
inputs: [
  {
    id: 'model-size',
    type: 'select',
    label: 'AI Model',
    options: [
      { value: 'llama3-8b', label: 'Llama 3 8B' },
      { value: 'llama3-70b', label: 'Llama 3 70B' }
    ],
    default: 'llama3-8b'
  },
  {
    id: 'quantization',
    type: 'select',
    label: 'Quantization',
    options: [
      { value: 'fp16', label: 'FP16' },
      { value: 'int8', label: 'INT8' },
      { value: 'int4', label: 'INT4' }
    ],
    default: 'fp16'
  }
],

result: {
  unit: 'GB VRAM',
  precision: 1
},

stats: [
  { id: 'stat-1', label: 'Min VRAM', key: 'min' },
  { id: 'stat-2', label: 'Recommended', key: 'rec' },
  { id: 'stat-3', label: 'GPU Tier', key: 'tier' },
  { id: 'stat-4', label: 'Status', key: 'status' }
]
};
```

Input types supported:

- number — Numeric input with min/max/step
- text — Text input
- select — Dropdown with options
- checkbox — True/False toggle

5B: calculate() Function

```
function calculate() {
  // Inputs read karo (auto-handled by template)
```

```

const values = {};
CONFIG.inputs.forEach(input => {
  const el = dom.('#' + input.id);
  if (!el) return;
  if (input.type === 'checkbox') {
    values[input.id] = el.checked;
  } else if (input.type === 'number') {
    values[input.id] = parseFloat(el.value) || 0;
  } else {
    values[input.id] = el.value;
  }
  state.inputs[input.id] = values[input.id];
});

// === YOUR CALCULATION LOGIC ===
// Yahan aapka formula ayega
// AI se bolo: "Mujhe VRAM calculation chahiye"
// AI yeh function likh dega

const model = values['model-size']; // e.g., 'llama3-8b'
const quant = values['quantization']; // e.g., 'fp16'

// Example logic (replace with real formula)
let vramNeeded = 0;
if (model === 'llama3-8b') vramNeeded = 16;
if (model === 'llama3-70b') vramNeeded = 140;

if (quant === 'int8') vramNeeded = vramNeeded / 2;
if (quant === 'int4') vramNeeded = vramNeeded / 4;

state.result = {
  value: vramNeeded,
  min: Math.ceil(vramNeeded * 0.9),
  rec: Math.ceil(vramNeeded * 1.2),
  tier: vramNeeded > 24 ? 'RTX 4090' : vramNeeded > 12 ? 'RTX 4070' : 'RTX 4060',
  status: vramNeeded > 24 ? 'High-end GPU needed' : 'Most GPUs supported'
};
// === END CALCULATION LOGIC ===

updateResults();
saveState();
}

```

5C: FAQ Questions

```

const items = [
  {
    question: 'How much VRAM for Llama 3 70B?',
    answer: '<p>Llama 3 70B requires approximately <strong>140GB VRAM</strong> at FP16 precision. With  

    ↪ INT4 quantization, this drops to around <strong>40GB</strong>.</p>'
  }
]

```

```
// ... aur questions add karo  
];
```

STEP 6: Edit data.json (Reference Data)

```
{  
  "last_updated": "2026-07-31",  
  "description": "VRAM requirements for popular AI models",  
  "items": [  
    {"name": "Llama 3 8B", "value": 16, "notes": "FP16 baseline"},  
    {"name": "Llama 3 70B", "value": 140, "notes": "FP16 baseline"},  
    {"name": "Mistral 7B", "value": 14, "notes": "FP16 baseline"}  
  ]  
}
```

data.json optional hai.

Agar aapka tool sirf user inputs pe based hai (jaise simple calculator), toh data.json hata bhi sakte ho. Tool.js mein loadData() automatically handle kar lega.

STEP 7: Register in tools-registry.json

File: shared/data/tools-registry.json

New tool add karo:

```
{  
  "slug": "vram",  
  "name": "VRAM Calculator for AI/LLMs",  
  "icon": "$$\vcenter{\hbox{\includegraphics[height=\ht\strutbox]{/usr/share/docminer/emoji/1f9e0.png}}}}$$",  
  "description": "Find out how much VRAM you need to run Llama, Mistral, Qwen, and other local AI models.",  
  "category": "AI",  
  "status": "live",  
  "tags": ["AI", "GPU", "LLM", "VRAM"],  
  "ogImage": "/assets/images/og-vram.jpg",  
  "schema": {  
    "applicationCategory": "UtilityApplication",  
    "keywords": ["vram calculator", "gpu memory", "ai model requirements"]  
  }  
}
```

Status values:

- "live" — Tool ready and working
- "coming-soon" — Tool under development
- "beta" — Tool live but experimental

Icon: Emoji use karo (🧠 ⚡ 💻 🔧 🔊)

STEP 8: Update sitemap.xml

File: sitemap.xml (root level)

Commented section mein se vram wala uncomment karo:

```
<!-- Before (commented) -->
<!--
<url>
  <loc>https://tools.redrag.in/vram/</loc>
  ...
</url>
-->

<!-- After (uncommented) -->
<url>
  <loc>https://tools.redrag.in/vram/</loc>
  <lastmod>2026-07-31</lastmod>
  <changefreq>weekly</changefreq>
  <priority>0.8</priority>
  <image:image>
    <image:loc>https://tools.redrag.in/assets/images/og-vram.jpg</image:loc>
    <image:title>VRAM Calculator</image:title>
    <image:caption>Calculate GPU memory for AI models</image:caption>
  </image:image>
</url>
```

STEP 9: Update robots.txt

File: robots.txt (root level)

Disallow line hatao:

```
# Before
Disallow: /vram/

# After (delete the line entirely)
```

TESTING CHECKLIST

Tool deploy ke baad yeh check karo:

- tools.redrag.in/vram/ open hota hai?
- Header aur footer dikh rahe hain?
- Inputs kaam kar rahe hain?
- Calculate button click pe result ata hai?
- Reset button kaam kar raha hai?
- Copy Results clipboard pe copy kar raha hai?

- Mobile pe responsive hai?
- FAQ expand/collapse ho raha hai?
- Console mein koi red error nahi?
- Hub page pe new tool card dikh raha hai?
- Google Rich Results Test pass kar raha hai?

QUICK REFERENCE: Common Tool Patterns

Pattern A: Simple Calculator (2-3 inputs)

Examples: RAM Latency, PPI Calculator

```
inputs: [  
  { id: 'freq', type: 'number', label: 'Frequency (MHz)', default: 3200 },  
  { id: 'cas', type: 'number', label: 'CAS Latency', default: 16 }  
]  
// Formula: (CAS / Frequency) * 2000 = Latency in ns
```

Pattern B: Lookup Table (select + data.json)

Examples: VRAM Calculator, PSU Calculator

```
inputs: [  
  { id: 'model', type: 'select', label: 'Select Model', options: [...] }  
]  
// Logic: Look up value from data.json based on selection
```

Pattern C: Multi-step Configurator

Examples: PC Build Configurator

```
inputs: [  
  { id: 'budget', type: 'select', label: 'Budget', options: [...] },  
  { id: 'use-case', type: 'select', label: 'Use Case', options: [...] },  
  { id: 'resolution', type: 'select', label: 'Resolution', options: [...] }  
]  
// Logic: Combination of selections determines recommendation
```

Pattern D: Comparison Tool

Examples: DDR4 vs DDR5, Resolution Scaling

```
inputs: [  
  { id: 'option-a', type: 'select', label: 'Option A', options: [...] },  
  { id: 'option-b', type: 'select', label: 'Option B', options: [...] }  
]  
// Logic: Calculate both, show side-by-side comparison
```

📞 NEED HELP?

Calculation logic nahi samajh aa raha?

AI se bolo: “*Mujhe [tool name] ka calculation logic chahiye. Inputs: [list]. Expected output: [description].*”

Schema validation fail ho raha hai?

Google Rich Results Test (search.google.com/test/rich-results) pe check karo.

Style break ho raha hai?

Browser console open karo (F12) → red errors check karo → screenshot leke AI ko bhejo.

Hub page pe tool nahi dikh raha?

`tools-registry.json` mein status "live" hai? JSON syntax valid hai? (trailing comma nahi honi chahiye last item pe)

🇮🇳 SCALABILITY: 100+ Tools Support

Yeh framework 100+ tools easily support karta hai because:

1. **Self-contained folders** — Har tool apne folder mein hai, koi overlap nahi
2. **Shared framework** — Common CSS/JS ek baar load hota hai, cache ho jata hai
3. **No build step** — GitHub Pages directly serve kar sakta hai
4. **Registry-driven hub** — Naya tool add karna = sirf JSON mein entry
5. **Lazy loading** — Har tool sirf jab visit ho tab load hota hai

GitHub Pages limits:

- Max 1GB repo size (100+ tools ke liye enough hai)
- 100GB bandwidth/month (plenty for traffic)
- Static files only (hamara framework exactly yahi hai)

Guide created: 2026-07-31

Version: 2.0.0 (Phase 8)

Next: Pick a tool from the priority list and start building!